HDOC DAY-46

Aim

To write a C program using a variable-length integer array and pointers to find the index of the smallest element greater than or equal to a given target. If the ceil value occurs more than once, the index of its first occurrence is returned.

Problem Statement

Read the size and elements of a sorted integer array in ascending order and an integer target. Pass the array to **ceil_index()** using its address. The function must find the smallest array element that is greater than or equal to the target and return its first index. Display the ceil element and its index, or display **-1** if no such element exists.

Algorithm

1. Start.
2. Read the size n of the array.
3. Read n sorted integer elements into arr[].
4. Read the target value.
5. Call ceil_index(arr, n, target) by passing the address of the array.
6. Inside ceil_index(), scan the array from index 0.
7. When the first element satisfying arr[i] >= target is found, return index i.
8. If no element satisfies the condition, return -1.
9. If the returned index is -1, display -1; otherwise display the ceil value and its index.
10. Stop.

C Program

```c
#include <stdio.h>

int ceil_index(int *arr, int n, int target)
{
    int i;

    for (i = 0; i < n; i++)
    {
        if (*(arr + i) >= target)
        {
            return i;
        }
    }

    return -1;
}

int main()
{
    int n, i, target, index;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d sorted elements: ", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter target: ");
    scanf("%d", &target);

    index = ceil_index(arr, n, target);

    if (index == -1)
    {
        printf("-1\n");
    }
    else
    {
        printf("Ceil of %d: %d at index %d\n",
               target, arr[index], index);
    }

    return 0;
}
```

Test Case Table

Test Case	Array	Target	Expected Output
1	5 7 7 8 8 10	6	Ceil of 6: 7 at index 1
2	5 7 7 8 9 10	8	Ceil of 8: 8 at index 3
3	5 7 7 8 8 10	11	-1

Output Snippets

Test Case 1

```
Enter the size of array: 6
Enter 6 sorted elements: 5 7 7 8 8 10
Enter target: 6
Ceil of 6: 7 at index 1
```

Test Case 2

```
Enter the size of array: 6
Enter 6 sorted elements: 5 7 7 8 9 10
Enter target: 8
Ceil of 8: 8 at index 3
```

Test Case 3

```
Enter the size of array: 6
Enter 6 sorted elements: 5 7 7 8 8 10
Enter target: 11
-1
```

Result

The program successfully finds the first index of the smallest element greater than or equal to the target using pointers. It displays -1 when no ceil element exists.